

Programming Homework: Linear Models with Scikit-learn

Problem 1: Comparing Linear Regression, Ridge, and LASSO

Requirements:

Using the `Diabetes` dataset from scikit-learn, perform the following tasks:

- a) Load and split the dataset into train/test sets (70/30 ratio).
- b) Train three models:
 - Linear Regression (ordinary linear regression)
 - Ridge Regression with $\alpha = 0.1$
 - LASSO Regression with $\alpha = 0.1$
- c) Evaluate performance using Mean Squared Error (MSE) on the test set.
- d) Compare the number of selected features (non-zero weights) for each model.
- e) Visualize and compare the weights of the three models using plots.
- f) Experiment with different α values and analyze their impact on performance.

Hints:

- Use `LinearRegression`, `Ridge`, and `Lasso` from `sklearn.linear_model`.
- Use `mean_squared_error` from `sklearn.metrics` for evaluation.
- Use `matplotlib.pyplot` to plot weight comparisons.
- Count non-zero weights using `np.sum(model.coef_ != 0)`.

Problem 2: Binary Classification with Logistic Regression

Requirements:

Using the `Breast Cancer` dataset from scikit-learn, perform the following tasks:

- a) Load the dataset and perform feature scaling using `StandardScaler`.
- b) Train a Logistic Regression model with different regularization strengths.
- c) Evaluate performance using metrics: accuracy, precision, recall, and F1-score.
- d) Plot the ROC curve and compute the Area Under the Curve (AUC).
- e) Analyze the effect of the regularization strength hyperparameter (C) on:
 - Model accuracy
 - Number of selected features (non-zero weights)
 - Values of the weights
- f) Compare performance when using L1 vs. L2 regularization.

Hints:

- Use `LogisticRegression` from `sklearn.linear_model`.
- Use `StandardScaler` from `sklearn.preprocessing` for feature scaling.
- Use `classification_report`, `roc_curve`, and `auc` from `sklearn.metrics`.
- Experiment with values of $C = [0.001, 0.01, 0.1, 1, 10, 100]$.
- Compare `penalty='l1'` and `penalty='l2'`.

Problem 3: Multi-tasking with Multi-output Regression

Requirements:

Using the `Linnerud` dataset from `scikit-learn` (containing multiple target variables), perform the following tasks:

- a) Load the dataset and explore the correlations among target variables.
- b) Train a multi-output regression model using:
 - Multi-output Linear Regression
 - Multi-output Ridge Regression
- c) Evaluate performance separately for each target variable.
- d) Compare results against training individual models for each target variable separately.
- e) Visualize predicted vs. actual values using scatter plots.

Hints:

- Use `MultiOutputRegressor` from `sklearn.multioutput`.
- Use `load_linnerud` to load the dataset.
- Evaluate using `r2_score` for each target variable.
- Compare with using standalone `LinearRegression` for each output individually.

References

- Scikit-learn Documentation: https://scikit-learn.org/stable/modules/linear_model.html
- Lecture Slides: Linear-Regression.pdf, Logistic-Regression.pdf, 2.MLP.pdf
- Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow